Lesson Overview: Streaming Big Data - Introduction

In the rapidly evolving data landscape, the ability to efficiently process and analyze real-time data is becoming increasingly crucial for businesses to remain competitive. For data engineers, mastering streaming data technologies and understanding the architecture of data pipelines are indispensable skills. This expertise enables the creation of flexible, scalable solutions that can handle the volume, velocity, and variety of big data, driving insights and innovation. In this lesson, we'll dive into advanced data handling techniques, exploring how batch and streaming data can be leveraged to transform business operations.



What You'll Learn:

Differentiate between batch and streaming data...

Differentiate between batch and streaming data, understanding their specific applications and business value.

Describe the principles of streaming...

Describe the principles of streaming technology and its impact on big data analytics.

Understand the common architectures...

Understand the common architectures, delivery guarantees, and challenges of streaming applications.

Ready to Dive In?

Reflect:

Think about the last time you watched a food delivery driver approach on a map, saw live sports scores update on an app, or noticed your social media feed refreshing with new posts instantly. These are all examples of streaming data in action.

Before you begin, can you think of one other daily activity that relies on up-to-the-second information? What would that experience be like if the data only updated once every hour?